

Lab Sheet 05

(01)

1. Loop inside a loop inside a loop that iterates until a given number $n - 1$ starting from 0 and prints all the indexes in between

$$O() = n * n * n$$

$$O(n^3)$$

2. First prints all the numbers from 1 to n then prints numbers from 1 to n^2

$$O() = n + n^2$$

$$O(n^2)$$

3. Loop inside a loop. Both runs from 0 to $n-1$. In each iteration sum is incremented by 1

$$O() = n * n + 1$$

$$O(n^2)$$

4. Prints the value of i from $i=n$ to $i=2$ while i is halving in each iteration

$$O() = \log n + 1$$

$$O(\log n)$$

(02)

a)

1. Take the first number.
2. Check whether it is equal to 300
3. If yes you have found the number.
4. If no check whether you have any numbers left in the array
5. If yes pick the next number and go to step 2
6. If no 300 is not in the array

b)

is 105 == 300?

is 210 == 300?

is 145 == 300?

is 320 == 300?

is 180 == 300?

is 250 == 300?

is 195 == 300?

is 410 == 300?

is 300 == 300?

- c) 9 comparisons are required
- d) best-case is the first number being the number 300. worst-case is the last number being the number 300 or number 300 not being in the array

(03)

- a)
 1. Take the middle number.
 2. Is it equal to 35.
 3. If yes you have found the number.
 4. If not is the number 35 greater than the middle number.
 5. If yes then set the new middle from the right part of the array
 6. If no then set the new middle from the left part of the array
 7. Is middle exists
 8. If yes continue from step 2 with the new middle
 9. If no the number 35 is not on the array

- b)

length of array = 10 | mid = $(10 // 2) = 5$ | is $25 == 35$?

length of array = 5 | mid = $(5 // 2) = 2$ | is $35 == 35$?

- c) 2 comparisons
- d) time complexity of binary search is $\log n$ since the array is divided in to half in each iteration

(04)

- a) loop iterates inside an array called posts. In each iteration an element inside the array is taken and its attribute value called likes is added to the totalLikes variable
- b) n times
- c) $n + 1 \rightarrow O(n)$
- d) since the big Oh notation is n, if the number of posts doubles the execution time will also doubles

(05)

- a) a loop inside a loop. Both loops runs from 0 to n - 1. The loop variables of first loop and the second loop are used to compare each and every student in the arrays student
- b) second loop runs n times and first loop runs n times.
therefore the total comparisons made is $= n * n = n^2$
- c) $O(n^2)$
- d) when the number of students increases the time complexity increased by a large amount since it uses the square of total number of students as n^2

(06)

- A) Algorithm C , Algorithm B , Algorithm A
- B) Algorithm A has a significant increase in no. of operations when compared with the no. of operations in Algorithm B
- C) Algorithm C, since the increment in n will have a very small affect to it compared to other 2 algorithms
- D) According to graph provided the change in the no. of operations of each algorithm when the value of n is small, is very small. But when n becomes larger the no. of operations in each algorithm changes drastically, so an algorithm that suited for a smaller dataset might not be suitable for a larger dataset since we cannot say for sure what its curve would look like from a smaller value for n .
- E) Algorithm C, if the value of n becomes very large, best algorithm to be used is C, since it has the lowest growth with n . Algorithm A and B will have an drastic increment in the no. of operations performed.